

Kit Four SKUs into a Bundle SOP (1x Episode: 4 SKUs)

One episode builds one bundle. Four cartons are loaded into one band sleeve in the order the kit card names them, one kit label is pressed onto the sleeve, the finished kit is set in the kit box, and the count is logged with one count tag. The table begins with the empty **band sleeve** on the **build spot** at the center, its open **mouth** facing right. A **SKU rack** stands along the right side with six **SKU bins** in one row, lettered **A** at the back to **F** at the front, each bin holding many of one carton type. The **kit card** stands in a card stand at the back-left and names the four SKUs to load, in load order. The **label pad** sits at the front-right corner, the **count tag rack** and the **log stand** sit at the front-center-right, and the empty **kit box** sits at the front-center.

The order never changes: load all four cartons, then label, then box, then log. No label is taken before all four cartons are home in the sleeve. The kit is not lifted before the label is pressed. The count tag does not go in the log stand until the kit is in the kit box.

The right gripper reads the kit card, picks every carton, pushes every carton home, takes and presses the kit label, lifts the finished kit into the kit box, and stands the count tag in the log stand. The left gripper presses the band sleeve flat on the build spot and holds it there while cartons go in and the label is pressed, then moves to the kit box and holds it down for the set-down. The left gripper works only at the center and the front-center. The right gripper works the right side, the center, and the front-right.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the band sleeve on the build spot, all six SKU bins on the rack, the kit card in its stand at the back-left, the label pad at the front-right corner, the count tag rack and log stand at the front-center-right, and the kit box at the front-center.
3. The build spot is visible from above, so the **label patch** on the sleeve's top panel, both sleeve mouths, and the gaps between cartons inside the sleeve can all be seen.
4. Both arms are at home with grippers open.

5. The table is bare apart from the band sleeve, the SKU rack, the kit card stand, the label pad, the count tag rack, the log stand, the kit box, and robot hardware.
6. The right arm reaches bin A, bin F, both sleeve mouths, the label pad, the tag rack, the log stand, and the inside of the kit box without stretching. The left arm reaches the left end of the band sleeve and the left rim of the kit box without stretching.

Materials checklist

1. One empty **band sleeve** sits on the **build spot** at the center of the table, **square** to the front edge, flat on the table, with its **mouth** facing right.
2. The sleeve is closed on the top, bottom, front, and back, and open at both ends. Inside, it is four cartons long and one carton wide, so four cartons fill it with no room left over.
3. The sleeve's top panel carries one printed **label patch**, centered, with no label on it.
4. The **SKU rack** stands along the right side of the table with six open-front **SKU bins** in one row, lettered **A** at the back to **F** at the front. The letters are printed on the front of each bin and are readable from above.
5. Each bin holds many **cartons** of one SKU only, and holds more than the kit card asks for. All six carton types are the same size and shape, so any of them fits the sleeve the same way. Each carton carries its own printed SKU label on one face.
6. The **kit card** stands upright in a card stand at the **back-left corner**, angled toward the front, and names four SKUs in load order: **A, C, D, F** — one carton each.
7. The **label pad** sits at the **front-right corner** and holds a liner of printed **kit labels**, printed face up. The top label's near edge is already lifted free of the liner.
8. The **count tag rack** sits at the **front-center-right** with upright **count tags** printed **1** through **6**, printing facing the front.
9. The **log stand** sits beside the tag rack with one empty **log slot**.
10. One empty **kit box** sits at the **front-center**, open on top, square to the front edge.
11. Before collection, confirm by hand that:
 - one carton can be pinched out of each bin without dragging others with it;
 - a carton slides into the sleeve mouth without the sleeve lifting or buckling;
 - four cartons fill the sleeve with no carton sticking out of either end;
 - a filled sleeve can be pinched at its middle and lifted without a carton sliding out;
 - a kit label comes away from the liner on a straight lift and sticks to the label patch;
 - a count tag stands on its own in the log slot.

Workspace layout

- **Build spot:** the center of the table — the band sleeve is loaded, labeled, and lifted from here
- **Right side:** the SKU rack, bins A to F, back to front

- **Back-left corner:** the kit card in its stand
- **Front-right corner:** the label pad
- **Front-center-right:** the count tag rack and the log stand
- **Front-center:** the kit box
- **Left side:** kept clear, so the left gripper can come in onto the sleeve and the kit box

Arm assignments

- **Left gripper:** presses the band sleeve flat on the build spot and holds it there through loading and labeling, then holds the kit box down by its left rim for the set-down.
- **Right gripper:** reads the kit card, picks cartons from the bins, pushes them home in the sleeve, takes and presses the kit label, lifts the kit into the kit box, and stands the count tag in the log stand.

Vocabulary

- **Build spot:** the place at the center of the table where the band sleeve sits. The sleeve stays here until the kit is lifted into the kit box.
- **Square:** the front face of the sleeve, or the front wall of the kit box, lines up with the front edge of the table, so neither end sits nearer the front.
- **Band sleeve:** the stiff open-ended sleeve the four cartons are loaded into. It is what bands the kit together, and it carries the kit label.
- **Mouth:** an open end of the band sleeve. The **right mouth** faces right and is the only end cartons go in through. The **left mouth** faces left and stays open.
- **Label patch:** the printed outline on the sleeve's top panel that the kit label is laid inside.
- **SKU:** one carton type. Each SKU lives in its own bin and has its own letter.
- **SKU rack:** the rack along the right side that holds the six bins.
- **SKU bin:** one open-front box on the rack, holding many cartons of one SKU only.
- **Bin letter:** the letter on the front of a bin, A at the back to F at the front.
- **Carton:** one item out of a bin. Four cartons make one kit.
- **Kit card:** the card at the back-left that names which four SKUs to load and in what order.
- **Load order:** the order the kit card lists the four SKUs, read top to bottom. The first SKU listed goes deepest into the sleeve.
- **Load:** slide one carton in through the right mouth with the right gripper.
- **Pushed home:** the carton has been slid in until it touches the carton already in front of it, or the far end of the sleeve if it is the first one, with no gap left.
- **Flush:** the last carton's right face is level with the right mouth, so no part of it sticks out.
- **Kit label:** one printed adhesive label from the label pad. One goes on each kit.

- **Lifted edge:** the near edge of the top label on the pad, already free of the liner. It is the only part of a label a gripper ever touches.
- **Press:** the closed right gripper set down flat on the laid label and held still for 2 seconds.
- **Seated:** the kit label lies inside the label patch, printing facing up, flat with no lifted corner and no bubble, and stays down when the right gripper comes away.
- **Kit:** the band sleeve with four cartons in it and the kit label pressed on.
- **Kit box:** the open-topped box at the front-center the finished kit is set in.
- **Count tag:** one of the printed number cards in the tag rack. One goes in the log stand each episode.
- **Count tag rack:** the holder at the front-center-right the count tags stand in.
- **Log stand:** the stand beside the tag rack with one **log slot**. Standing a count tag in it is how the count is logged.
- **Standing:** the count tag is pushed down into the log slot, stands upright with its printing facing the front, and stays there when the right gripper lets go.

Steps

Run Steps 1–5 in order on the one band sleeve, then end the episode with Step 6.

Step 1: Hold the band sleeve down

Goal: the band sleeve is pinned flat on the build spot and cannot slide while cartons go in.

- With the **left gripper**, press down on the **left end** of the sleeve's top panel and hold it against the table.
- Keep the **left gripper** there through Steps 2 and 3.

Check: the sleeve is flat on the table, **square** to the front edge, its **mouth** faces right, and it does not slide when the left gripper presses. If the sleeve is crooked, straighten it with the **left gripper** first, then press it down.

Expected state: the sleeve is held, both mouths are open, the sleeve is empty, and the label patch is bare.

Step 2: Load the four SKUs from the kit card

Goal: four cartons sit in the sleeve, in the order the kit card names, all pushed home.

- With the **left gripper**, keep pressing the sleeve down.
- With the **right gripper**, move **one carton at a time**. Never scoop, never carry two cartons, and never tip a bin out.
- Keep every carry low over the table. Do not lift a carton high above the sleeve.

2.1 Read the kit card

- With the **right gripper** clear of the sleeve, read the **kit card** and note the next SKU in **load order**.
- Read the card again for every carton. Do not work from memory.

2.2 Pick one carton

- With the **right gripper**, reach into the bin with that letter and pinch **one carton**.
- Lift it straight up and clear of the bin lip, then carry it left to the sleeve's **right mouth**.
- Turn it so its printed SKU label faces **up**. Every carton goes in the same way up.
- If more than one carton comes out together, put the extra cartons back in that same bin with the **right gripper** before going on.

2.3 Push the carton home

- With the **right gripper**, set the carton into the **right mouth** so its left face is inside the sleeve.
- Push it straight left along the table until it is **pushed home** — against the far end for the first carton, against the carton already in for the rest.
- Open the **right gripper** and come away without pulling the carton back.
- Repeat 2.1 to 2.3 until all four SKUs on the card are in.

Check: four cartons are in, in the order the card names them, with no gap between any two, and the last carton's right face is **flush** with the right mouth. If a carton sits short, push it home with the **right gripper**. If a carton sticks out past the mouth, push it in. If a carton is the wrong SKU or the wrong way up, pull it and everything loaded after it back out through the **right mouth** with the **right gripper**, put the wrong carton back in its own bin, and load again from the card.

Expected state: the sleeve holds four cartons, tight end to end, every SKU label facing up, the label patch still bare, the six bins still on the rack, and the kit card still standing in its stand.

Step 3: Press the kit label on

Goal: one kit label is seated inside the label patch on the sleeve's top panel.

- With the **left gripper**, keep pressing the sleeve down.
- With the **right gripper**, close on the **lifted edge** of the top label on the **label pad** and lift straight up, so the label comes away from the liner.
- Carry the label in one go to the sleeve's top panel, sticky face down. Do not set it down on the way and do not let it touch the table.
- Lower it straight down into the **label patch**, printing facing up, with its edges in line with the patch, then open the **right gripper**.

- Close the **right gripper** and set it down flat on the label. **Press** for 2 seconds, then lift straight up.

Check: the label is **seated** — inside the patch, printing up, flat with no lifted corner and no bubble, and it stays down when the right gripper comes away. If a corner lifts, press that corner again with the closed **right gripper** for 2 seconds. If the label landed outside the patch or upside down, leave it where it is and press a second label into the patch on top of it.

Expected state: the sleeve holds four cartons and carries one seated kit label, and the label pad still lies flat at the front-right corner.

Step 4: Box the kit

Goal: the finished kit lies in the kit box, label facing up.

- With the **left gripper**, let go of the sleeve and press down on the **left rim** of the **kit box**.
- With the **right gripper**, pinch the **band sleeve** at its middle, front face and back face, so the jaws hold the sleeve and not a carton.
- Lift the kit straight up, just clear of the build spot. Keep it level.
- Carry it forward to the **kit box** and lower it in until it rests on the bottom, label facing up and the sleeve lying the same way round as it did on the build spot.
- Open the **right gripper** and come away.
- Keep the **left gripper** on the box rim until the right gripper is clear.

Check: the kit lies flat in the box with all four cartons still in the sleeve and the label facing up. If a carton slides out during the lift, set the kit back on the build spot with the **right gripper**, push the carton home through the **right mouth**, and lift again.

Expected state: the kit box holds one finished kit, the build spot is bare, and the label pad, tag rack, and log stand are untouched.

Step 5: Log the count

Goal: one count tag stands in the log slot showing how many cartons went into the kit.

- With the **left gripper**, come off the kit box and hold clear of the front-center.
- With the **right gripper**, pinch the count tag printed **4** by its top edge and lift it straight up out of the **count tag rack**.
- Carry it to the **log stand** and line it up with the **log slot**, printing facing the front.
- Push it straight down into the slot until it stands on its own, then open the **right gripper**.

Check: one tag is **standing** — upright in the log slot, printing **4** facing the front, and it stays there when the right gripper lets go. If the tag leans, falls, or faces backwards, take it out with the **right gripper**, turn it printing-forward, and push it down again.

Expected state: the kit box holds one labeled kit, one tag printed 4 stands in the log slot, and the tag rack still holds its other tags upright.

Step 6: End the episode

Goal: recording ends with the kit built, labeled, boxed, and logged.

1. Confirm the end state:
 - the kit box holds one kit with four cartons in the band sleeve;
 - the four SKUs are the ones the kit card names, in card order, every SKU label facing up;
 - one kit label is **seated** in the label patch, printing facing up;
 - the build spot is bare and no carton is loose on the table;
 - one count tag printed **4** stands in the log slot;
 - all six bins are still on the rack and the kit card is still standing in its stand.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Take the count tag out of the log slot and stand it back in the tag rack, printing facing the front. Leave the log slot empty.
2. Lift the kit out of the kit box. Peel the kit label off the sleeve and throw it away.
3. Slide the four cartons out of the sleeve and put each one back in its own bin, A into A through F into F. Do not mix SKUs between bins.
4. Pick up any carton that ended up on the table or the floor and put it back in its own bin.
5. Set an empty band sleeve **square** on the build spot, flat on the table, mouth facing right, with a bare label patch. Use a fresh sleeve if the used one is dented, torn, or no longer holds four cartons snug.
6. Lift the near edge of the new top label on the label pad free of the liner.
7. Set the empty kit box square at the front-center, and stand the kit card upright in its stand at the back-left, readable from the front.
8. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the cartons and the fixtures. Throw out any carton that is crushed, split, or has a torn SKU label. Top each bin back up so it holds more cartons than the kit card asks for. Replace a sleeve that is dented or soft, replace a count tag that is bent or torn, and replace the kit

card if it is creased or no longer readable. Fit a fresh liner on the label pad if it is running low.

2. Leave one empty sleeve square on the build spot, the six bins full on the rack in letter order, the kit card upright in its stand, a fresh label lifted on the label pad, six count tags standing in the tag rack, the log slot empty, and the kit box empty and square at the front-center.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Violation: Band sleeve not held

- **Visible cue:** the left gripper is off the sleeve while the right gripper pushes a carton in or presses the label, and the sleeve slides, lifts, buckles, or is chased across the build spot.
- **SOP rule broken:** Step 1 (the left gripper presses the sleeve down by its left end and holds it there through Steps 2 and 3).
- **Coaching note:** left gripper on the sleeve first, and leave it there until the label is pressed.

Violation: Band sleeve moved off the build spot

- **Visible cue:** either gripper turns, drags, or lifts the sleeve during loading or labeling, the sleeve ends Step 3 out of square with the front edge, or its mouth no longer faces right.
- **SOP rule broken:** Steps 1–3 (the sleeve stays flat and square on the build spot, mouth facing right, until the kit is lifted in Step 4).
- **Coaching note:** bring the cartons and the label to the sleeve; never move the sleeve to them.

Violation: Kit card not read, moved, or covered

- **Visible cue:** either gripper knocks, moves, or lays the card flat, something is set on top of it, or the right gripper loads carton after carton without the card ever coming back into view.
- **SOP rule broken:** Step 2.1 (read the kit card before every carton; do not work from memory, and leave the card standing in its stand).
- **Coaching note:** the card is the pick list, not your memory. Read it, and leave it standing where it is.

Violation: Wrong SKU loaded and left in

- **Visible cue:** the right gripper reaches into a bin whose letter is not on the kit card, or a wrong SKU is seen in the sleeve and the right gripper moves on to the label instead of pulling it and the cartons loaded after it back out.
- **SOP rule broken:** Step 2.2 (reach into the bin with the letter the card names) and Step 2 (pull a wrong carton and everything loaded after it back out through the right mouth, return it to its own bin, and load again from the card).
- **Coaching note:** say the bin letter out loud before you reach in. Fix it before you label it — a labeled kit with the wrong SKU in it cannot be undone.

Violation: SKUs loaded out of card order

- **Visible cue:** the right gripper loads the four SKUs in some order other than the order the card lists, so the deepest carton in the sleeve is not the first SKU on the card.
- **SOP rule broken:** Step 2 (load in the order the kit card names, first listed deepest).
- **Coaching note:** top of the card goes in first and travels furthest. Work down the list.

Violation: More than one carton moved at a time, or a bin tipped out

- **Visible cue:** the right gripper scoops, rakes, or lifts two or more cartons together, tips a bin to pour cartons out, or carries a handful to the sleeve, and the extras are not put back.
- **SOP rule broken:** Step 2 (move one carton at a time; if more than one comes out together, put the extras back in that same bin before going on).
- **Coaching note:** one pinch, one carton. If two come up, drop the spare back in the bin first.

Violation: Carton loaded the wrong way up

- **Visible cue:** a carton goes into the sleeve with its printed SKU label facing down, sideways, or to the back, so the cartons in the sleeve are not all the same way up.
- **SOP rule broken:** Step 2.2 (turn each carton so its printed SKU label faces up; every carton goes in the same way up).
- **Coaching note:** label up, every time. Turn the carton before it reaches the mouth, not inside the sleeve.

Violation: Carton not pushed home

- **Visible cue:** a gap is left between two cartons, a carton sticks out past the right mouth, or the right gripper pulls a carton back toward the mouth as it comes away, and the episode moves on.
- **SOP rule broken:** Step 2.3 (push each carton straight left until it is pushed home, and come away without pulling it back; the last carton's right face is flush with the right mouth).
- **Coaching note:** push until it stops, then open the gripper and lift away — do not drag it back.

Violation: Carton dropped in or carried high

- **Visible cue:** the right gripper opens above the sleeve and the carton falls or bounces in, or it lifts a carton high above the table instead of a low carry to the mouth.
- **SOP rule broken:** Steps 2.2 and 2.3 (keep every carry low over the table, and set the carton into the right mouth before pushing it home).
- **Coaching note:** low and level to the mouth, then slide. Nothing goes in from above.

Violation: Wrong number of cartons in the sleeve

- **Visible cue:** Step 3 starts with three or fewer cartons in the sleeve, or with five in, or the right gripper picks twice from the same bin so one SKU on the card is missing.
- **SOP rule broken:** Step 2 (all four SKUs the card names are in the sleeve, one carton each, before the label is taken).
- **Coaching note:** count the cartons in the sleeve against the card before you go to the label pad.

Violation: Bin lifted, dragged, or spilled

- **Visible cue:** either gripper lifts a bin, drags it along the rack, tips it, or leaves cartons spilled on the rack or the table.
- **SOP rule broken:** Step 2.2 (reach into the bin; the bins stay on the rack in letter order for the whole episode).
- **Coaching note:** reach into the bin, never move the bin. Pinch cartons out one by one.

Violation: Kit left unlabeled

- **Visible cue:** the episode reaches Step 4 or ends with the label patch bare, or with a label lying loose on the sleeve, in the box, or on the table instead of pressed into the patch.
- **SOP rule broken:** Step 3 (one kit label is seated in the label patch before the kit is lifted).
- **Coaching note:** no kit leaves the build spot without its label. Label, then lift.

Violation: Kit label not seated

- **Visible cue:** the label sits outside the patch, crooked to it, upside down, or with a lifted corner or a bubble, or the right gripper never presses it flat for a full 2 seconds.
- **SOP rule broken:** Step 3 (lower the label into the patch printing up with its edges in line, then press flat for 2 seconds; press a lifted corner again).
- **Coaching note:** line it up before you let go, then hold the press for a full two seconds.

Violation: Label mishandled on the way

- **Visible cue:** the right gripper takes more than one label off the liner, touches the sticky face, folds or creases the label, sets it down on the way, or lets it touch the table.
- **SOP rule broken:** Step 3 (close on the lifted edge only, and carry the label in one go to the sleeve, sticky face down).
- **Coaching note:** lifted edge only, one label, straight to the patch.

Violation: Kit lifted by a carton, or the kit came apart in the lift

- **Visible cue:** the right gripper closes on a carton rather than on the sleeve, tilts the kit during the lift, or a carton slides out on the way to the box and the episode moves on without it being pushed back home.
- **SOP rule broken:** Step 4 (pinch the band sleeve at its middle, front face and back face, lift level; if a carton slides out, set the kit back on the build spot, push it home, and lift again).
- **Coaching note:** grip the sleeve, keep it level, and go slowly. If one slides out, go back and refit it.

Violation: Kit not boxed, or boxed wrong

- **Visible cue:** the episode ends with the kit still on the build spot, resting on the box rim, leaning against the outside of the box, or lying on the table beside it; or the kit lies in the box with its label facing down or sideways, or turned end for end from how it lay on the build spot.
- **SOP rule broken:** Step 4 (lower the kit into the kit box until it rests on the bottom, label facing up, the sleeve the same way round as it lay on the build spot).
- **Coaching note:** carry it level, do not turn your wrist, and take it all the way down to the bottom of the box before you open the gripper.

Violation: Kit box not held, or moved off its spot

- **Visible cue:** the left gripper is off the box rim while the right gripper lowers the kit in, or either gripper drags, lifts, or knocks the box off the front-center spot.
- **SOP rule broken:** Step 4 (the left gripper presses the box down by its left rim and stays there until the right gripper is clear).
- **Coaching note:** hold the box before the kit comes over it, and let go only after the gripper is out.

Violation: Count not logged

- **Visible cue:** the episode ends with the log slot empty, or with a count tag lying loose on the table, in the kit box, or still in the tag rack.
- **SOP rule broken:** Step 5 (one count tag is pushed down into the log slot before the episode ends).
- **Coaching note:** the log is the last thing you fill. No kit is finished until the count is in.

Violation: Wrong count tag used

- **Visible cue:** a tag printed with a number other than **4** stands in the log slot, or more than one tag is put in the slot.
- **SOP rule broken:** Step 5 (pinch the count tag printed 4 and stand it in the log slot).
- **Coaching note:** four cartons, tag four. Read the number on the tag before you lift it.

Violation: Count tag not standing

- **Visible cue:** the tag leans, falls over, sits on top of the log slot, or stands with its printing facing away from the front, and the right gripper leaves it that way.
- **SOP rule broken:** Step 5 (push the tag straight down into the log slot printing facing the front, and let go only once it stands on its own).
- **Coaching note:** printing to the front, push it down, then let go and watch that it stays.

Violation: Fixture knocked out of place

- **Visible cue:** either gripper tips, pushes, or drags the SKU rack, the kit card stand, the label pad, the count tag rack, or the log stand off its spot, spills the tags, or lifts the liner off the pad.
- **SOP rule broken:** Steps 2, 3, and 5 (the bins stay on the rack, the card stays standing in its stand, and the label pad, tag rack, and log stand stay on their spots through the whole episode).
- **Coaching note:** reach in over the lip. Work slower and lower so nothing gets shoved.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence — the right gripper goes to the label pad before all four cartons are in, lifts the kit before the label is pressed, takes a count tag before the kit is in the box, or goes back to load a carton after the label is on.
- **SOP rule broken:** Steps 2–5 (load all four, label, box, then log, in that order).
- **Coaching note:** finish the phase before you start the next one. Load, label, box, log.

Violation: Carton, label, or tag dropped and left

- **Visible cue:** either gripper drops a carton, a kit label, or a count tag onto the rack, the table, or the floor, and the episode moves on without it being picked up.

- **SOP rule broken:** Steps 2–5 (a carton that ends up on the table is picked up with the right gripper and put back in its own bin, and a dropped label or tag is retrieved before the step goes on).
- **Coaching note:** pick up what you drop before the next pick. Nothing stays loose on the table.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper picking a carton, pushing a carton home, handling a label or a tag, or lifting the kit, or the right gripper holding down the sleeve or the kit box.
- **SOP rule broken:** Steps 1–5 (the right gripper picks, pushes, labels, lifts, and logs; the left gripper holds the sleeve, then the kit box).
- **Coaching note:** right gripper does the work, left gripper holds. It does not swap.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the kit, the label, the box, and the logged tag are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 6 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A bin arrives empty, so a SKU the card names cannot be picked.
- A bin arrives holding two SKUs mixed together.
- A carton is crushed or split, so it will not slide into the sleeve.
- The band sleeve is torn or dented, so four cartons will not sit snug in it.
- A kit label will not come off the liner, or has no adhesive left, so it cannot be seated.
- The log slot is broken or spread open, so a correctly pushed count tag will not stand.
- The kit card is misprinted or unreadable, so the SKUs cannot be read.

Annotation subtasks (from SOP)

1. Press the band sleeve flat on the build spot and hold it with the left gripper

2. Read the kit card for the next SKU in load order
3. Pick one carton from that bin with the right gripper
4. Turn the carton label up and set it into the right mouth
5. Push the carton home along the sleeve with the right gripper
6. Repeat the read, pick, and push for all four SKUs on the card
7. Take one kit label from the label pad by its lifted edge
8. Lay the label in the label patch and press it for 2 seconds
9. Move the left gripper to the kit box rim
10. Lift the kit by the band sleeve with the right gripper
11. Lower the kit into the kit box, label facing up
12. Take the count tag printed 4 from the tag rack
13. Stand the count tag in the log slot, printing facing the front
14. Confirm the end state, return both arms home, and end the episode